

BEYOND NODAL DAMPING

WHEN STABLE REPLACEMENTS FAIL TOGETHER

Minimal Incompatibility Hypergraphs for Safe SG → IBR Transition Planning

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1 MOTIVATION & QUESTION

Replacing SG units changes the full portfolio. A stable endpoint can still contain an unstable intermediate coalition.

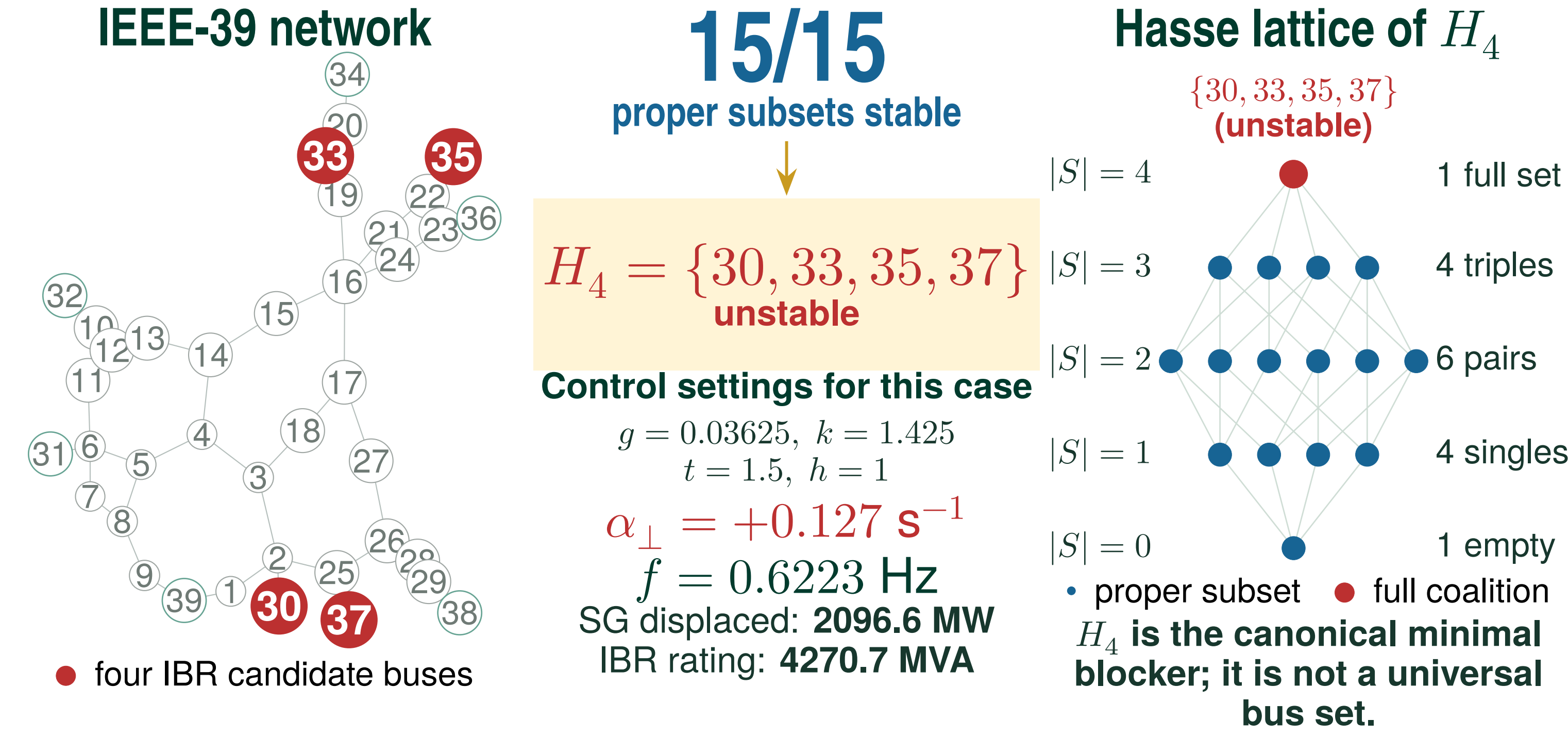


Which replacement combinations are incompatible, why do they fail, and how can paths avoid them?

$$\mathcal{H}(\theta) = \text{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Edges mark minimal unstable coalitions.

2 FLAGSHIP RESULT: IEEE-39



3 STABLE TARGET ≠ SAFE PATH

Three nested guarantees for a target portfolio T :

A FINAL STATE STABLE $\alpha_{\perp}(T, \theta) < 0$	B SOME ORDER IS SAFE Stable prefix chain exists	C EVERY ORDER SAFE No blocker in T .
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$C \Rightarrow B \Rightarrow A$
Converse implications can fail.
 $C(T) \iff \forall e \in \mathcal{H}(\theta), e \not\subseteq T$

Recovery of exact blocker sets Matches across 52 operating points		
first-order modal		22/52
additive		31/52
pairwise		38/52

15/15
smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

+0.127 s⁻¹
critical growth rate
canonical portfolio

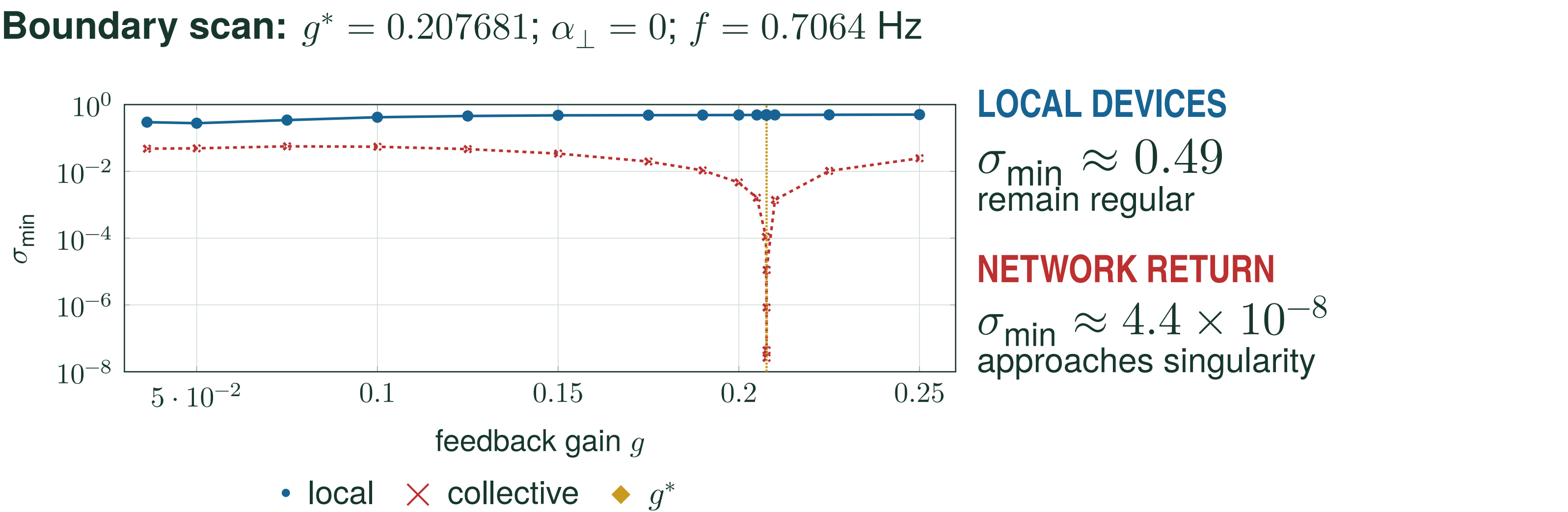
0.6223 Hz
target-band frequency
nominal IEEE-39 case

4A FROM DEVICES TO NETWORK RETURN

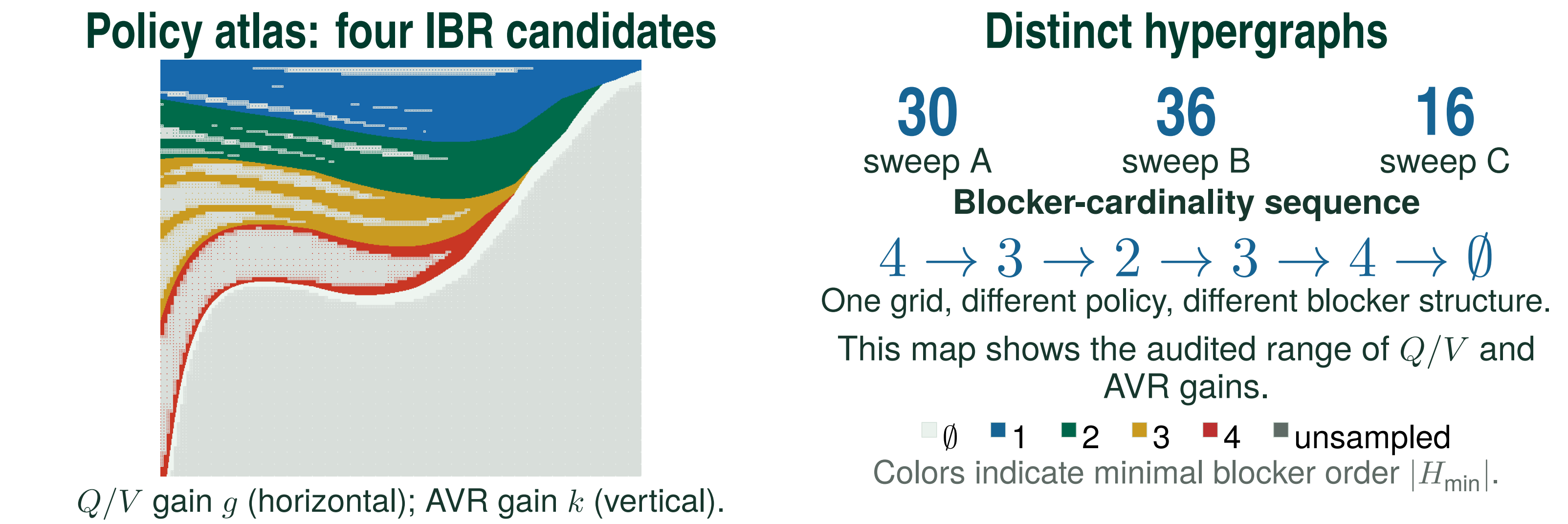
Network update	$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$
Characteristic	$\det P_S(s) = h_S(s) \det T_0(s) \det(I + M_{SS}(s))$
Local × collective	$\det(I + M_{SS}) = \left[\prod_{i \in S} \det(I + M_{ii}) \right] \det(I + Q_{SS})$
Interactions	$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$

A one-mode reduction remains unverified at the strict residual tolerance.

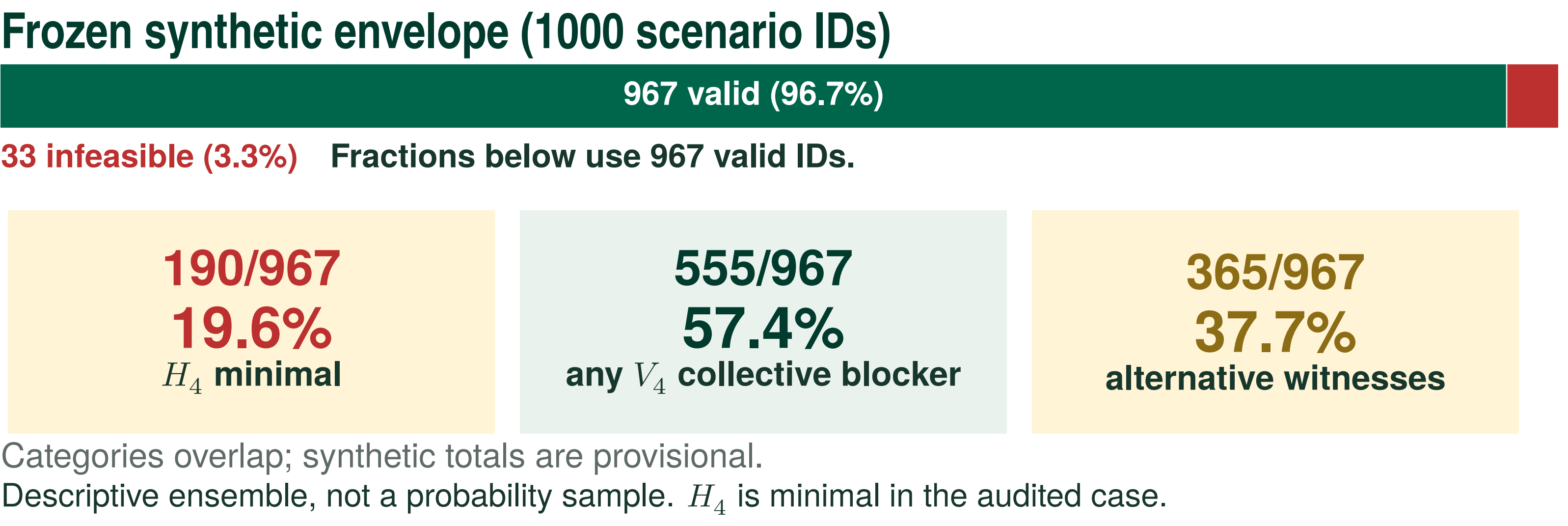
4B THE COLLECTIVE TERM CLOSES



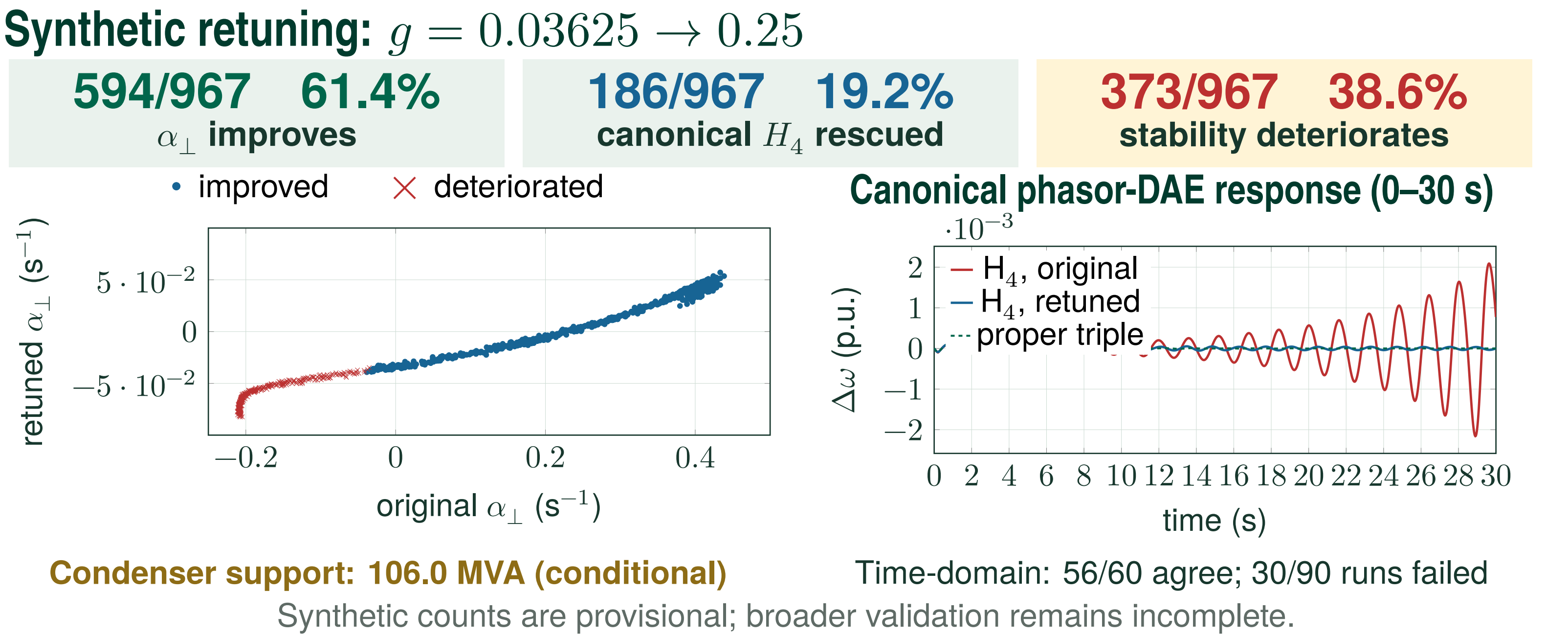
5 POLICY CHANGES THE MINIMAL BLOCKER



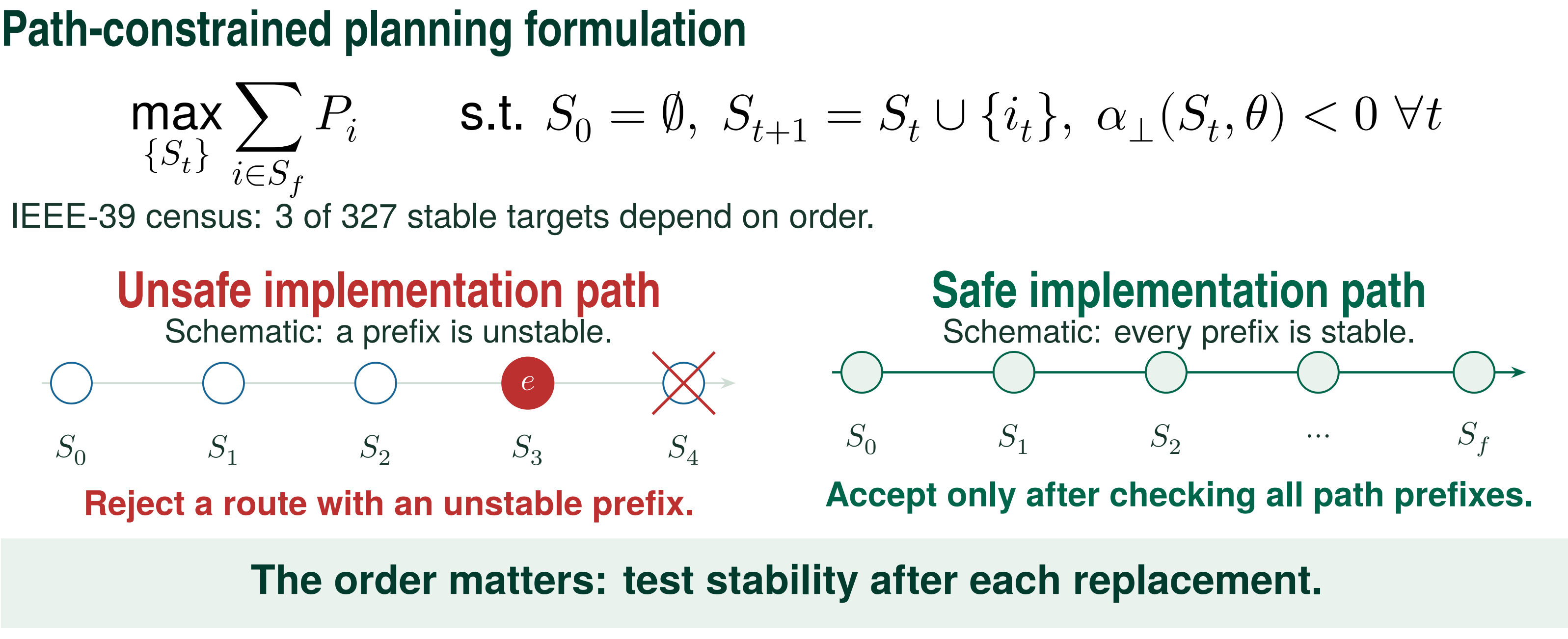
6 THE CANONICAL H_4 IS NOT UNIVERSAL



7 A CANONICAL REPAIR IS NOT UNIVERSAL



8 SAFE PATHS CHECK EVERY PREFIX



9A MODEL CHECKS



9B WHAT “SAFE PATH” MEANS HERE

Each prefix is a **re-equilibrated phasor-DAE** portfolio.
The analysis does not simulate **physical switching**, EMT, current limits or DC-link limits.
Blockers depend on the model and control policy.

TAKEAWAY

SAFE PREFIXES, SAFE TRANSITIONS.

A stable endpoint can still hide an unstable intermediate coalition.
Blockers change with the control policy.
Safety here refers to re-equilibrated phasor-DAE prefixes.

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